

SHABBONA 5 NNE, IL, USA

WMO: 724970

Lat: 41.8430N

Lon: 88.8513W

Elev: 262

StdP: 98.21

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 54811

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.5	-18.8	-25.2	0.4	-21.7	-22.5	0.5	-18.5	10.8	-6.2	9.7	-8.5	3.5	N/A	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.5	23.1	30.1	22.6	28.7	21.9	25.1	29.6	24.1	28.2	23.1	27.4	2.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.2	28.2	22.7	18.0	27.1	21.7	16.9	26.1	78.1	29.7	73.5	28.1	70.0	27.6	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.7	6.7	-25.7	33.5	3.8	1.4	-28.4	34.5	-30.6	35.3	-32.7	36.1	-35.5	37.1		
			-25.7	26.7	4.1	0.8	-28.6	27.2	-31.0	27.7	-33.3	28.1	-36.2	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-6.2	-4.8	2.7	9.0	15.5	20.6	21.7	20.4	17.3	10.5	3.3	-3.2
	DBStd	11.18	6.95	6.74	6.23	5.04	4.97	3.36	3.15	2.90	4.28	5.17	5.43	6.23
	HDD10.0	1921	502	414	242	78	9	0	0	2	57	208	409	
	HDD18.3	3764	761	647	487	282	118	16	7	13	69	248	451	667
	CDD10.0	1544	0	1	14	48	178	317	362	323	220	73	8	1
	CDD18.3	345	0	0	1	2	29	82	111	78	38	6	0	0
	CDH23.3	3130	0	0	7	35	296	732	995	637	365	63	0	0
	CDH26.7	847	0	0	0	2	84	221	288	143	98	11	0	0
Wind	WSAvg	2.8	3.5	3.6	3.5	3.5	3.0	2.4	1.7	1.5	1.8	2.4	2.9	3.2
Precipitation	PrecAvg	878	47	40	55	80	113	102	94	99	72	71	64	41
	PrecMax	1102	83	131	105	137	204	239	183	256	154	168	138	77
	PrecMin	551	13	12	22	38	21	26	26	15	11	5	9	
	PrecStd	128	21	27	28	33	55	57	44	49	42	46	39	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	15.0	24.4	26.8	31.7	32.9	33.9	31.8	31.7	28.4	20.1	14.4
		MCWB	9.9	10.8	17.3	16.6	20.5	22.7	25.2	23.5	22.5	21.2	14.1	13.2
	2%	DB	7.6	9.8	18.4	24.0	29.0	31.0	31.2	29.8	29.7	25.1	16.4	9.9
		MCWB	5.9	7.4	13.8	15.4	19.7	22.5	24.5	23.5	21.9	17.1	11.8	7.9
	5%	DB	4.4	5.7	15.2	21.2	26.8	29.4	29.6	28.3	27.3	22.0	14.0	7.1
		MCWB	3.0	3.9	11.4	13.8	18.3	21.5	23.3	22.4	20.6	16.1	11.0	5.7
	10%	DB	2.4	3.2	12.0	18.5	24.4	27.5	28.2	26.8	25.4	19.5	11.6	4.8
		MCWB	1.4	1.8	8.8	12.3	17.5	20.9	22.3	21.3	19.5	15.2	8.4	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	12.3	18.0	18.5	22.7	25.4	26.7	25.6	24.5	21.0	15.7	13.8
		MCDB	11.2	14.6	24.3	23.4	29.1	30.0	31.5	29.5	29.4	25.1	18.3	14.8
	2%	WB	5.5	8.8	14.9	16.2	21.3	23.9	25.6	24.3	23.1	18.8	13.0	9.2
		MCDB	7.2	11.4	18.4	21.6	27.3	28.5	30.3	28.0	28.0	21.2	15.7	10.3
	5%	WB	2.7	4.8	12.3	14.6	20.1	22.9	24.6	23.3	21.8	17.1	10.8	6.1
		MCDB	3.9	6.4	15.0	19.4	25.7	27.7	28.9	27.1	26.1	20.4	13.8	7.5
	10%	WB	1.3	2.0	8.7	12.9	18.9	21.8	23.5	22.2	20.5	15.4	8.5	3.8
		MCDB	2.2	3.3	11.6	17.4	23.5	26.2	27.5	25.7	23.9	18.9	11.0	5.0
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.3	9.3	12.0	11.6	11.0	11.0	11.2	12.8	11.5	9.3	7.4
		MCDBR	9.1	10.8	13.3	16.3	14.3	12.8	11.8	12.4	14.2	14.8	12.6	9.7
		MCWBR	8.1	8.7	9.0	9.5	7.1	6.6	6.5	7.1	6.5	8.0	8.8	8.4
	5% WB	MCDBR	8.7	11.6	12.3	13.8	11.9	10.9	10.5	10.8	11.1	11.4	11.6	9.6
	MCWBR	7.6	9.1	8.5	9.4	6.5	6.4	6.5	6.6	5.9	8.3	8.8	8.5	
Clear-Sky Solar Irradiance	taub	0.285	0.303	0.334	0.387	0.415	0.424	0.437	0.421	0.394	0.342	0.301	0.283	
	taud	2.416	2.380	2.403	2.265	2.238	2.257	2.208	2.248	2.299	2.449	2.509	2.449	
	Ebn at Noon	868	905	914	882	862	853	836	838	836	849	847	840	
	Edh at Noon	78	95	105	130	137	135	140	131	116	88	71	70	
All-Sky Solar Radiation	RadAvg	1.68	2.62	3.64	4.65	5.36	6.09	6.08	5.35	4.48	2.94	1.90	1.40	
	RadStd	0.19	0.33	0.31	0.39	0.40	0.45	0.34	0.32	0.38	0.34	0.17	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page